

P08. Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection

中英双语博士精读笔记

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Theme / 主题: Data-driven predictive control + mixed traffic safety

Method family / 方法族: Data-driven predictive control + RLS / 数据驱动预测控制与递归最小二乘

PDF pages / 页数: 6 **Relevance / 相关度:** 78/100

Source / 来源: <https://arxiv.org/abs/2203.05739>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Data-driven predictive control + mixed traffic safety”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Data-driven predictive control + mixed traffic safety. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Signalized and rear-end-focused rather than conflict-zone scheduling across all movements.

1.2 核心科学问题 / Core Scientific Question

在混合交通信号交叉口中，如何在线估计前车行为并使 CAV 在黄灯/红灯附近安全预测控制？

How can a CAV estimate a preceding HDV online and perform safety-aware predictive control near signal changes?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Finite-horizon predictive control uses recursive least squares to estimate preceding HDV behavior in real time.
- 安全/避碰贡献 (safety contribution): Prioritizes rear-end collision avoidance when preceding human-driven vehicles approach signal changes.
- 平滑/停止避免贡献 (smoothing contribution): Smooths CAV response near yellow/red phases while avoiding overly conservative braking behind HDVs.
- 创新力度评判 (reviewer judgement): 中等: 安全问题具体、方法稳健, 对混合交通 fallback 很有参考价值。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Data-driven predictive control + RLS	数据驱动预测控制与递归最小二乘	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?

- 能否复述实验基准是否公平，指标是否覆盖效率、安全、能耗和计算时间？
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验？

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection^{*}

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Abstract: A typical urban signalized intersection poses significant modeling and control challenges in a mixed traffic environment consisting of connected automated vehicles (CAVs) and human-driven vehicles (HDVs). In this paper, we address the problem of deriving safe trajectories for CAVs in a mixed traffic environment that prioritizes rear-end collision avoidance when the preceding HDVs approach the yellow and red signal phases of the intersection. We present a predictive control framework that employs a recursive least squares algorithm to approximate in real time the driving behavior of the preceding HDVs and then uses this approximation to derive safety-aware trajectory in a finite horizon. We validate the effectiveness of our proposed framework through numerical simulation and analyze the robustness of the control framework.

Keywords: Connected automated vehicles; predictive control; data-driven parameter estimation; vehicle safety; mixed traffic environment.

1. INTRODUCTION

Optimal coordination of connected automated vehicles (CAVs) can improve network performance, e.g., fuel economy, traffic throughput, at traffic scenarios such as urban intersections, see Talebpour and Mahmassani (2016). In recent efforts, a decentralized optimal control framework has been established for real-time coordination of CAVs traveling through signal-free automated intersections, see Malikopoulos et al. (2021); Chalaki and Malikopoulos (2021); Mahbub and Malikopoulos (2021); Kumaravel et al. (2021). However, these approaches have been developed based on the strict assumption of 100% CAV penetration rate which is currently not realizable; see Alessandrini et al. (2015).

CAVs must be able to safely co-exist with human-driven vehicles (HDVs) resulting in a *mixed traffic environment*, which pose significant modeling and control challenges due to the stochastic nature of human-driving behavior. Furthermore, the use of conventional traffic lights is still the most prevalent way of traffic control at urban intersections that adds an extra layer of complexity in modeling the HDV behavior due to the presence of the zone of yellow light dilemma; see Zhang et al. (2014). Thus, the need for an efficient CAV control framework considering the inclusion and interaction of HDVs approaching the signalized intersections is essential to provide safety assurance under unknown HDV behavior.

Several research efforts have adopted adaptive cruise control (ACC) for automated vehicles in a mixed traffic envi-

ronment to tackle the HDV behavior and ensure rear-end collision avoidance; see Jiang et al. (2007); Yuan et al. (2009). Lu et al. (2019) considered a variation of the car-following model to design an eco ACC controller. However, ACC controllers using car-following models such as the intelligent driver model (IDM) (Treiber and Kesting, 2013) do not always perform well since they can have stability implications leading to rear-end collision; see Milanés and Shladover (2014). Milanés et al. (2013) proposed a cooperative adaptive cruise controller where the control parameters are derived using system identification on real-world experimental data. However, such control parameters cannot capture the instantaneous changes in HDV behavior. Naus et al. (2010) proposed an explicit model predictive control (MPC) ACC controller that employs a prediction model with a constant speed assumption of the preceding vehicle and does not consider the complex car-following dynamics of the human driver. Dollar et al. (2021) utilized an IDM model to identify offline the human driving styles in a car-following scenario and developed an MPC-based cruise control for a CAV. Jin and Orosz (2018) proposed an optimal cruise control design in which feedback gains and driver reaction time of HDVs were estimated in real time by a sweeping least square method.

In this paper, we consider the problem related to controlling a CAV while approaching a signalized intersection in the presence of multiple preceding HDVs with unknown driving behavior. To generate safe and optimal control actions for the CAV, we propose a data-driven predictive control framework that takes into account the future trajectories of the HDVs to ensure that the collision does not take place over a finite-time horizon. The control framework is then implemented in a receding horizon manner for

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图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

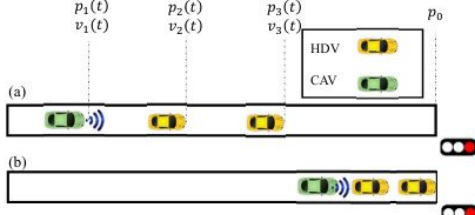


Fig. 1. A connected and automated vehicle (green) and human-driven vehicles (yellow) approaching a signalized intersection at red signal phase.

robustness against stochastic driving behavior of HDVs. The constant time headway relative velocity (CTH-RV) model and the recursive least squares (RLS) algorithm are utilized to estimate the HDV's driving behavior given the data collected online. We evaluate the efficiency of the proposed method by numerical simulations that employ a nonlinear car-following model to replicate the human drivers.

The remainder of the paper is organized as follows. In Section 2, we present the modeling framework and formulate the problem. In Section 3, we provide a detailed exposition of the safety-aware and data-driven predictive control framework with real-time behavior estimation. In Section 4, we evaluate the effectiveness of the proposed approach in a simulation environment. Finally, we draw conclusions and discuss the future research directions in Section 5.

2. PROBLEM FORMULATION

We consider multiple HDVs followed by a CAV traveling on a single-lane road and approaching an urban signalized intersection with a red (or yellow then red) traffic signal phase (Fig. 1). Note that, the general idea of our formulation can be extended to different cases such as yield/stop traffic sign, downstream traffic congestion, and pedestrian crossing, where the preceding HDVs' motion can change abruptly to come to a full stop. Next, to facilitate our exposition we provide the following definitions.

Definition 1. Suppose that the red signal phase is active at some time instant $t = t^0$. The set $\mathcal{N}$ of the vehicles approaching the intersection at $t = t^0$ is $\mathcal{N} = \{N, N-1, \dots, 1\}$, where $N \in \mathbb{N}$ is the total number of vehicles under consideration. Here, the vehicles are assigned integer indices by the order of their respective distances from some fixed stopping position p_0 located downstream near the signal head. The indices $N, N-1, \dots, 2$ represent the HDVs followed by the CAV denoted by the index 1.

Definition 2. The set of HDVs at time instant $t = t^0$ is $\mathcal{N}_{\text{HDV}} = \mathcal{N} \setminus \{1\}$.

When the red signal phase is active at some time instant $t = t^0$, the HDV- N in $\mathcal{N}$ must stop behind the position p_0 . The objective of the CAV-1 is to derive an optimal trajectory to stop behind HDV-2 such that no rear-end collision takes place.

Remark 3. In our formulation, we require that the set of HDVs $\mathcal{N}_{\text{HDV}}$ is non-empty at time instant $t = t^0$ when the red signal phase of the intersection is active. If $\mathcal{N}_{\text{HDV}}$ is empty, then the problem of avoiding rear-end collision becomes redundant.

2.1 Communication Topology

The CAV-1 is retrofitted with appropriate sensors and communication devices to estimate in real time the state information of the preceding HDVs in $\mathcal{N}_{\text{HDV}}$ through vehicle-to-everything communication protocol and intelligent roadside units; see Duan and Yang (2021). We refer to the unidirectional flow of information from the preceding HDVs in $\mathcal{N}_{\text{HDV}}$ to the trailing CAV-1 as the multi-predecessor communication topology.

We impose the following assumption.

Assumption 4. Communication to and from the CAV occurs without any delays and errors.

Assumption 4 may be strong, but it is relatively straightforward to relax it as long as the noise in the measurements and/or delays is bounded.

In contrast, the trajectory of each HDV i in $\mathcal{N}_{\text{HDV}}$ is solely dictated by the perception of the state information of the immediate preceding HDV $i+1$ in $\mathcal{N}_{\text{HDV}}$. For the leading vehicle HDV- N that does not have a preceding vehicle, its driving actions depend on the relative distance to the stopping point.

2.2 Vehicle Dynamics and Constraints

We consider the following discrete-time model with a sampling time $\tau \in \mathbb{R}^+$ to represent the dynamics of each vehicle $i \in \mathcal{N}$,

$$p_i(t+1) = p_i(t) + v_i(t)\tau + \frac{1}{2}u_i(t)\tau^2, \quad (1a)$$

$$v_i(t+1) = v_i(t) + u_i(t)\tau, \quad (1b)$$

where $p_i(t) \in \mathcal{P}_i$, $v_i(t) \in \mathcal{V}_i$, and $u_i(t) \in \mathcal{U}_i$ denote the position, speed and control input (acceleration/deceleration) of each vehicle i in $\mathcal{N}$. The sets $\mathcal{P}_i$, $\mathcal{V}_i$, and $\mathcal{U}_i$, $i \in \mathcal{N}(t)$, are complete and totally bounded subsets of $\mathbb{R}$. Note that in the discrete-time dynamics model (1), we assume that the control input $u_i(t)$ of each vehicle i in $\mathcal{N}$ remains constant in the time period of length τ between time instants t and $t+1$, which is different to some previous approaches that assume constant speed between time instants t and $t+1$; see Naus et al. (2010); Kianfar et al. (2012).

To ensure that the control input and vehicle speed are within a given admissible range, the following constraints are imposed,

$$u_{\min} \leq u_i(t) \leq u_{\max}, \quad \text{and} \quad (2a)$$

$$0 \leq v_{\min} \leq v_i(t) \leq v_{\max}, \quad (2b)$$

where $u_{\min}$, $u_{\max}$ are the maximum braking and acceleration, respectively, of each vehicle in $\mathcal{N}$, and $v_{\min}$, $v_{\max}$ are the minimum and maximum speed limits, respectively.

The control input $u_i(t)$ of each vehicle $i \in \mathcal{N}$ in (1) can take different forms based on the consideration of connectivity and automation. For CAV-1 in $\mathcal{N}$, we consider a switching control framework based on the following

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

cases: if at time instant $t = t^0$ (Remark 3) (a) $\mathcal{N}_{\text{HDV}}$ is empty, then CAV-1 derives its control input by using its default adaptive cruise controller (see Milanés and Shladover (2014)). (b) if $\mathcal{N}_{\text{HDV}}$ is not empty, then CAV-1 derives and implements the control input $u_1(t)$ using the proposed control framework discussed in Section 3.

For each HDV $i \in \mathcal{N}_{\text{HDV}}$, however, we consider a car-following model to represent the predecessor-follower coupled dynamics (Fig. 1) with its preceding vehicle $i+1$ that has the following generic structure

$$u_i(t) = f_i(\Delta p_i(t), v_i(t), \Delta v_i(t)), \quad (3)$$

where $f_i(\cdot)$ represents the behavioral function of the car-following model of vehicle $i \in \mathcal{N}_{\text{HDV}}$, and $\Delta p_i(t) := p_{i+1}(t) - p_i(t) - l_c$ and $\Delta v_i(t) := v_{i+1}(t) - v_i(t)$ denote the headway and approach rate of vehicle i with respect to its preceding vehicle $i+1$, respectively. We consider two edge cases that may arise from the above definitions: (a) if there is no vehicle $i+1$ preceding vehicle i within a certain look-ahead distance d_f , then we consider $\Delta p_i(t) = d_f$ and $\Delta v_i(t) = 0$, and (b) if there is an obstruction/red signal phase immediately ahead of vehicle i at a distance d_s , then $\Delta p_i(t) = d_s$ and $\Delta v_i(t) = -v_i(t)$. There are several car-following models reported in the literature that can emulate a varied class of human driving behavior; see Weng and Wu (2001).

The parameters of a car-following model can be recovered from historical data using offline identification methods; see Treiber and Kesting (2013). However, since the historical data might not be available and the human driving behavior usually changes over time, offline identification methods do not work well in practice. As a result, in our proposed framework, we consider that the CAV does not have full prior knowledge of the behavioral function $f_i(\cdot)$ of the preceding HDVs. Instead, the CAV assumes a specific type of car-following model for the HDV, then estimates the model parameters for each HDV online using real-time collected data. A method for estimating car-following model parameters of the HDVs is given in Section 3.1.

To capture the car-following characteristics of the preceding HDV-2's dynamics from the CAV-1's control point of view, we define additional states as

$$e_p(t) = p_2(t) - p_1(t) - l_c, \quad (4a)$$

$$e_v(t) = v_2(t) - v_1(t). \quad (4b)$$

To introduce the rear-end collision avoidance constraint, we first use the following definition of dynamic safe following headway $s_i(t)$.

Definition 5. The dynamic safe following headway $s_i(t)$ between two consecutive vehicles i and $(i+1) \in \mathcal{N}$ is

$$s_i(t) = \rho_i v_i(t) + s_0, \quad (5)$$

where $\rho_i \in \mathbb{R}^+$ denotes a desired time headway that each vehicle $i \in \mathcal{N}$ maintains while following the preceding vehicle, and $s_0 \in \mathbb{R}^+$ is the standstill distance denoting the minimum bumper-to-bumper gap at stop.

The rear-end collision avoidance constraint between CAV-1 and its immediately preceding HDV-2 can thus be written as

$$e_p(t) \geq s_1(t). \quad (6)$$

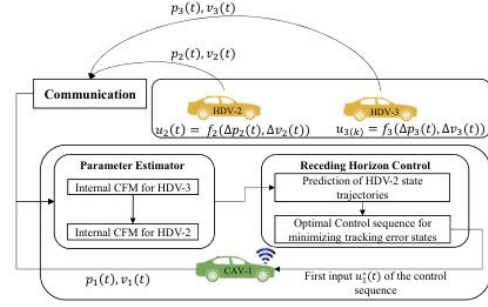


Fig. 2. The structure of the proposed control framework to address Problem 6.

We now formalize the main objective of the CAV-1 control framework.

Problem 6. Given the multi-predecessor communication topology (Section 2.1), the main objective of CAV-1 is to derive its optimal control input $u_1^*(t)$ such that CAV-1 adapts to its preceding HDV's driving behavior in real time and drives the states $e_p(t)$ and $e_v(t)$ to their respective reference states with minimum control effort satisfying the state, control and safety constraints in (2)-(6).

3. CONTROL FRAMEWORK

In our approach, we adopt a receding horizon predictive control framework with multi-predecessor communication topology and data-driven estimation of HDVs' car-following parameters for state prediction to address Problem 6, as shown in Figure 2. In the receding horizon control, the optimal control input at the current time step is obtained by solving a predictive control problem with a horizon T_p while only the first element of the obtained control input sequence is implemented. Afterward, the horizon moves forward one step, and the above process is repeated until a final horizon is reached; see Borrelli et al. (2017). Note that, the prediction horizon T_p is usually selected empirically to best accommodate the control performance and computational requirement. The essential steps of the proposed framework are outlined as follows.

- (1) **Data-driven parameter estimation:** At each time instant t , the current states $p_i(t), v_i(t)$ of each preceding HDV i in $\mathcal{N}_{\text{HDV}}$ is communicated to CAV-1. Since the exact car-following model f_i of each HDV i in $\mathcal{N}_{\text{HDV}}$ is unknown to CAV-1, it considers a specific type of car-following model to represent the driving behavior of each HDV, and estimates the parameters of the car-following model for each HDV online.
- (2) **Predictive control problem:** CAV-1 then uses the estimated car-following model from Step 1 to predict the future state trajectories of the immediately preceding HDV-2 and derives its own optimal control input sequence $U_1^*(t) := [u_1^*(t), u_1^*(t+1), \dots, u_1^*(t+T_p-1)]^T$ using the receding horizon control framework discussed above. Finally, CAV-1 implements only the first control input $u_1^*(t)$.

图 3: 核心方法或算法页 / Core method or algorithm page (p.3, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

In what follows, we provide a detailed exposition of the steps discussed above.

3.1 Online Car-following Model Parameter Estimation

In this section, we use a recursive least-squared formulation (Ljung and Söderström, 1983) to estimate the parameters of the internal car-following model residing in CAV-1's mainframe to represent the driving behavior of each of the preceding HDVs. To this end, we consider the CTH-RV model (Wang et al., 2020)

$$v_i(t+1) = v_i(t) + \eta_i(\Delta p_i(t) - \rho_i v_i(t))\tau + \nu_i(v_i(t) - v_i(t))\tau, \quad (7)$$

where the model parameters η_i and ν_i are the control gains on the constant time headway and the approach rate, and ρ_i is the desired safe time headway for each HDV i in $\mathcal{N}_{\text{HDV}}$, respectively. We employ the linear CTH-RV model instead of other complex nonlinear models so that the resulting control problem presented in the next section is thus convex and can be solved efficiently in real-time. Moreover, it is also observed that the CTH-RV model is highly comparable to other nonlinear car-following models in terms of data fitting (Gunter et al., 2019).

Suppose that we measure the speed $v_i(t)$, headway gap $\Delta p_i(t)$ and approach rate $\Delta v_i(t)$ for each preceding HDV i in $\mathcal{N}_{\text{HDV}}$ with sampling rate τ . We recast (7) as

$$v_i(t+1) = \gamma_{i,1}v_i(t) + \gamma_{i,2}\Delta p_i(t) + \gamma_{i,3}v_i(t), \quad (8)$$

where $\gamma_{i,1} := (1 - (\eta_i\rho_i + \nu_i)\tau)$, $\gamma_{i,2} := \eta_i\tau$ and $\gamma_{i,3} := \nu_i\tau$ are the parameters that can be estimated using the RLS algorithm. The original model parameters η_i, ν_i and ρ_i are then uniquely determined from $\gamma_{i,1}, \gamma_{i,2}, \gamma_{i,3}$ as long as $\gamma_{i,2} \neq 0$. Next, we can write (8) in matrix form as

$$v_i(t+1) = \gamma_i^T \phi_i(t), \quad (9)$$

where $\phi_i(t) := [v_i(t), \Delta p_i(t), v_i(t)]^T$ is the regressor vector and $\gamma_i := [\gamma_{i,1}, \gamma_{i,2}, \gamma_{i,3}]^T$ is the parameter vector. We can estimate γ_i using the following recursive least squares algorithm as follows (Ljung and Söderström, 1983)

$$\hat{\gamma}_i(t) = \hat{\gamma}_i(t-1) + L_i(t)[v_i(t) - \hat{v}_i(t)], \quad (10a)$$

$$\hat{v}_i(t) = \hat{\gamma}_i^T(t-1)\phi_i(t), \quad (10b)$$

$$L_i(t) = \frac{P_i(t-1)\phi_i(t)}{\xi + \phi_i^T(t)P_i(t-1)\phi_i(t)}, \quad (10c)$$

$$P_i(t) = \frac{1}{\xi} \left[P_i(t-1) - \frac{P_i(t-1)\phi_i(t)\phi_i^T(t)P_i(t-1)}{\xi + \phi_i^T(t)P_i(t-1)\phi_i(t)} \right]. \quad (10d)$$

Here, $\xi \in [0, 1]$ is the forgetting factor that assigns a higher weight to the recently collected data points and discounts older measurements, and $\hat{\gamma}_i(t)$ denotes the estimate of the parameter vector γ_i at time instant t , which is updated recursively as new data becomes available. In what follows, we introduce the predictive control problem that is needed to be solved.

3.2 Predictive Control Problem

The main objective of the predictive controller of the CAV is to (a) drive the position tracking state $e_p(t)$ to a reference $e_{p,r}(t)$, (b) drive the speed tracking state $e_v(t)$ to zero, and (c) minimize CAV-1's control input $u_1(t)$. To this end, the receding horizon controller generates

the predictive states $e_p(t+n|t), e_v(t+n|t)$ for $n = 1, \dots, T_p$ at each time instant t for a predictive horizon T_p using the state definitions in (4), vehicle dynamics in (1) and internal car-following models of the HDVs in (7) approximated in the previous section. Then the control input sequence $U_1(t) := [u_1(t), u_1(t+1), \dots, u_1(t+T_p-1)]^T$ is derived such that the predictive states are driven to their respective reference states. The predictive control problem thus can be written as

$$\min_{U_1(t)} \frac{1}{2} \sum_{n=1}^{T_p} \left[w_{e_p}(e_p(t+n|t) - e_{p,r}(t+n|t))^2 + w_{e_v}e_v(t+n|t)^2 + w_u(u_1(t+n-1))^2 \right], \quad (11)$$

subject to :

model: (1), (3), (4),

constraints: (2), (6),

reference state: $e_{p,r}(t) := s_1(t)$,

where the predictive reference state $e_{p,r}(t+n|t)$ can be computed using the relation $e_{p,r}(t) = s_1(t)$ and the dynamics model in (1) and (4), and $w_{e_p}, w_{e_v}, w_u \in \mathbb{R}^+$ are the weights on the reference tracking of the headway $e_p(t)$, speed deviation $e_v(t)$, and the CAV-1's control input $u_1(t)$, respectively.

The predictive control problem in (11) can be transformed into a standard constrained quadratic programming problem and solved using commercially available solvers; see Andersson et al. (2019a). At each discrete time instant t , the optimal control sequence $U_1^*(t)$ is computed by solving (11) and only the first control input $u_1^*(t)$ is applied. Then the system moves to the next time instant $t+1$ and the process is repeated until a final time horizon is reached.

Remark 7. While implementing the above control framework, if any of the preceding HDVs leaves the current lane or passes the intersection at any time instant t , we simply update the sets $\mathcal{N}$ and $\mathcal{N}_{\text{HDV}}$ starting from the next time instant $t+1$, where the control problem (11) is again solved with the updated information.

4. SIMULATION AND RESULT

This section validates the performance of the proposed safety-aware data-driven predictive control by numerical simulations at a mixed-traffic signalized intersection.

4.1 Simulation Setup

In the simulations, we utilize a nonlinear car-following model namely the optimal velocity model (OVM) to generate the driving actions of simulated human drivers (Bando et al., 1995). The car-following OVM is given as

$$u_i(t) = \alpha(V_i(t) - v_i(t)) + \beta\Delta v_i(t), \quad (12)$$

$$V_i(t) = \frac{v_d}{2} (\tanh(\Delta p_i(t) - s_i(t)) + \tanh(s_i(t))).$$

The parameters of the OVM for each HDV include the driver's sensitivity coefficients α and β , and the desired speed v_d . These parameters for the simulated HDVs are assumed to be different to each other and chosen by random perturbations up to 20% around the following nominal values: $\alpha = 0.8$, $\beta = 0.6$, $v_d = 15.0$ m/s, $\rho =$

图 4: 实验或结果页 / Experiments or results page (p.4, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

Table 1. Parameters of the controller

Parameters	Value	Parameters	Value
τ	0.1 s	T_p	50
$v_{\max}$	15 m/s	$v_{\min}$	0 m/s
$u_{\max}$	3 m/s ²	$u_{\min}$	-5 m/s ²
ρ	2.0 s	s_0	3.0 m
w_{e_p}	1	w_{e_v}	0.1
w_u	1		

2.0 s, $s_0 = 5.0$ m. The parameters and weights in the predictive control framework used for the simulations are given in Table 1. The RLS-based estimators are initialized with the following values: $\gamma_i(0) = [0.67, 0.1, 0.18]^T$ and $P_i(0) = 0.01 \mathbb{I}_3$ where $\mathbb{I}_3$ is the 3×3 identity matrix, while the forgetting factor is chosen as $\xi = 1.0$. The impact of ξ on RLS algorithm is investigated in detail by Vahidi et al. (2005) and thus, omitted here. Python is used in the simulations in which the constrained optimal control problem is formulated by CasADi framework; see Andersson et al. (2019b), and solved by the built-in qpOASES solver.

4.2 Results and Discussions

The results for a numerical simulation involving a CAV and 2 preceding HDVs are illustrated in Fig. 3, in which the longitudinal positions, speeds, and headways of all the vehicles are given in Figures 3a, 3b and 3c, respectively. Note that the position by which the vehicles must stop is $p_0 = 0$, and for the leading HDV, the headway is computed as the relative distance to the stopping point. As can be seen from Figures 3a-3c, the simulated HDVs slow down and then stop while approaching the signalized intersection. Given the behavior of the HDVs, the proposed control framework can perform safe and comfortable braking for the CAV without violating any of the state, input, and safety constraints.

Moreover, to assess the scalability of the proposed control framework to the number of preceding vehicles, we conduct three other simulations for the scenarios with 4, 5, and 6 vehicles (3, 4, and 5 HDVs, respectively) and illustrate the vehicle trajectories in Fig. 4. These results verify that the proposed control framework works effectively with different numbers of preceding vehicles.

Finally, the estimated parameters in the CTH-RV car-following model for the HDV-2 are depicted in Fig. 5. As more real-time data are added to update the estimations, the car-following parameters stabilize to the set of values that accurately describes the driving behavior of the HDVs. Therefore, using the linear CTH-RV model and online RLS technique, we can approximate a nonlinear car-following model and use this estimation to predict the future states of the HDVs.

5. CONCLUDING REMARKS

In this paper, we addressed the problem of a CAV traveling in a mixed traffic environment and approaching a signalized intersection. A data-driven predictive control framework was developed in which the car-following behavior of HDVs ahead of the CAV is modeled by the CTH-RV model with online estimated parameters through the RLS

algorithm. In the proposed framework, by utilizing data-driven car-following models, the CAVs can predict the future behavior of the HDVs and then derive their optimal safety-aware trajectory in a finite horizon. The proposed control framework was validated by numerical simulations with multiple preceding HDVs showing that the generated control actions can ensure safe braking for the CAVs. A direction for future research should focus on extending this framework to consider multi-lane traffic intersections with lane changing behavior of the HDVs.

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图 5: 结论或局限页 / Conclusion or limitations page (p.5, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 1. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Data-driven predictive control + mixed traffic safety: 既要提高效率, 又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法, 还是可以由更强的优化器/更保守规则解决?

p.2 2. PROBLEM FORMULATION

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口, 应与公式“RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立?

p.2 2.1 Communication Topology

章节目标 / Learning goal: 承接前后章节, 补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献, 但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

p.2 2.2 Vehicle Dynamics and Constraints

章节目标 / Learning goal: 承接前后章节, 补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献, 但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

p.3 3. CONTROL FRAMEWORK

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 控制器用 RLS 实时估计前方 HDV 行为，再在有限时域内求解预测控制问题，重点避免黄灯/红灯切换时的追尾风险和过度保守制动。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.4 3.1 Online Car-following Model Parameter Estimation

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口，应与公式“RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.4 3.2 Predictive Control Problem

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

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审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.4 4. SIMULATION AND RESULT

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环：Numerical simulation and robustness analysis.

技术焦点 / Technical focus: 重点追踪指标：最小车间距、碰撞风险、速度/加速度平滑性、延误和信号违规。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：在混合交通信号交叉口中，如何在线估计前车行为并使 CAV 在黄灯/红灯附近安全预测控制？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：前车行为可由低维参数模型描述。；CAV 可可靠观测前车状态。；主要风险是纵向追尾，而非多方向冲突区碰撞。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Finite-horizon predictive control uses recursive least squares to estimate preceding HDV behavior in real time., 输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Numerical simulation and robustness analysis.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC
3	核心求解 / Core solver	Finite-horizon predictive control uses recursive least squares to estimate preceding HDV behavior in real time.
4	安全接口 / Safety interface	Prioritizes rear-end collision avoidance when preceding human-driven vehicles approach signal changes.
5	平滑与效率 / Smoothing and efficiency	Smooths CAV response near yellow/red phases while avoiding overly conservative braking behind HDVs.
6	验证 / Validation	Numerical simulation and robustness analysis.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	3/5	RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC
算法贡献 / Algorithmic contribution	5/5	Finite-horizon predictive control uses recursive least squares to estimate preceding HDV behavior in real time.
实验贡献 / Experimental contribution	3/5	Numerical simulation and robustness analysis.
工程贡献 / Engineering contribution	4/5	Prioritizes rear-end collision avoidance when preceding human-driven vehicles approach signal changes.
博士课题价值 / PhD relevance	4/5	Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

RLS 估计与有限时域 MPC / RLS estimation and finite-horizon MPC

$$\hat{\theta}_k = \hat{\theta}_{k-1} + K_k(y_k - \phi_k^\top \hat{\theta}_{k-1}), \quad \min_{u_{0:N-1}} \sum_{k=0}^N \ell(x_k, u_k, \hat{\theta}_k)$$

Symbol / 符号	含义	Meaning	Role / 建模作用
theta	HDV 行为参数	HDV behavior parameters	在线估计前车模型
K_k	RLS 增益	RLS gain	控制新观测对参数的影响
phi_k	回归特征	regressor features	由前车状态构成
N	预测时域	prediction horizon	MPC 规划长度
ell	阶段代价	stage cost	综合安全、舒适和信号响应

5.2 核心算法架构 / Core Algorithm Layout

控制器用 RLS 实时估计前方 HDV 行为, 再在有限时域内求解预测控制问题, 重点避免黄灯/红灯切换时的追尾风险和过度保守制动。

The controller estimates preceding HDV behavior with RLS and solves a finite-horizon predictive-control problem focused on rear-end safety near signal transitions.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

MPC 的滚动优化提供反馈修正, RLS 提供模型自适应; 但稳定性依赖可辨识性、预测时域和安全约束设置。

Receding-horizon MPC offers feedback correction and RLS adapts the behavior model, but robustness depends on identifiability and constraint design.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 前车行为可由低维参数模型描述。
- CAV 可可靠观测前车状态。
- 主要风险是纵向追尾, 而非多方向冲突区碰撞。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境: 固定参数 MPC、普通跟驰模型、无估计控制器和保守刹车策略。

Validation / 验证: Numerical simulation and robustness analysis.

Metrics / 指标: 最小车间距、碰撞风险、速度/加速度平滑性、延误和信号违规。

Ablation / 消融: 关键是去掉 RLS 自适应、改变预测时域、改变安全约束强度, 观察鲁棒性。

Reviewer reading / 审稿式阅读提醒: 阅读实验时不要只看平均值; 应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 11 figure references and 2 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 场景局限于信号交叉口纵向跟驰。
- RLS 对异常驾驶和非线性行为变化可能响应不足。
- 未解决交叉冲突、合流和多车调度。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把 RLS/贝叶斯预测接入无信号 JSSP smoother，为 HDV 建立不确定处理时间。
- 用分布鲁棒 MPC 替代点估计 MPC。
- 将追尾 CBF 与横向冲突 CBF 统一成多约束 safety layer。

7.3 如何服务当前课题 / Use in This Project

Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 预测时域 / Prediction horizon, range [2, 12] s.

时域越长越能提前制动，但计算量和模型误差暴露也增加。

A longer horizon anticipates braking but increases computation and model-error exposure.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator, 右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block, 紧跟符号表。
- 审稿人批注用 reviewer-note block, 博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线, 打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	1. INTRODUCTION
p.2	2. PROBLEM FORMULATION
p.4	4. SIMULATION AND RESULT

p.1 1. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。 / Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 2. PROBLEM FORMULATION 把交通交互转写为状态、约束、目标或图结构, 是精读时最应复现的部分。 / Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.2 2.1 Communication Topology 作为局部论证段落阅读, 关注它如何服务主问题。 / Read as a local argument and ask how it supports the main question.

p.2 2.2 Vehicle Dynamics and Constraints 作为局部论证段落阅读, 关注它如何服务主问题。 / Read as a local argument and ask how it supports the main question.

p.3 3. CONTROL FRAMEWORK 给出算法流程和求解机制, 应重点追踪输入、输出和安全接口。 / Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.4 3.1 Online Car-following Model Parameter Estimation 把交通交互转写为状态、约束、目标或图结构, 是精读时最应复现的部分。 / Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 3.2 Predictive Control Problem 把交通交互转写为状态、约束、目标或图结构, 是精读时最应复现的部分。 / Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 4. SIMULATION AND RESULT 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。 / Tests efficiency, safety, or energy claims; inspect whether baselines are fair.